

Static and Dynamic Ring Assignment in a Clock-Forwarded DWDM Optical Link

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Abstract We present heater sweep-based in-situ minimum-power assignment of microrings to lasers during clock-forwarded optical link initialization (“static ring assignment”). We also show rotation of microring-to-laser assignment during live link operation (“dynamic ring assignment”). ©2026 The Authors

Introduction

AI systems require increasingly tightly connected compute to continue to advance in capability and scale. The resulting link reach and bandwidth demands are now being addressed by introducing optical links into the AI cluster scale-up domain [1]. Integrating the optical engine (OE) in these links on the same interposer as the host helps avoid bandwidth bottlenecks at the host electrical interface [2]. Ring resonator-based clock-forwarded dense WDM (DWDM) is a natural architecture for these OEs: ring compactness and clock forwarding-enabled jitter tracking yield bandwidth-dense and energy-efficient links [3].

In ring-based WDM systems, rings must first be assigned to lasers before link operation may begin, and this must be done with only the resources available on the chip. In this work, we present an *in-situ* method for this static ring assignment (SRA) step that automatically finds the lowest-power configuration and aligns TX and RX rings to lasers in a way that is more robust to link design, thermal crosstalk, and self-heating than prior approaches. We also show RX-side dynamic ring assignment (DRA), in which the RX ring-to-laser alignment is rotated during live link operation to reduce heater power.

Test Chip Architecture and Thermal Tuning

We demonstrate SRA and DRA on a 9λ 32 Gb/s/λ clock-forwarded optical link in a 3D-stacked 7 nm electronic IC (EIC)/65 nm photonic IC (PIC) SiPh platform (Fig. 1a) [4,5]. The EIC contains analog front-ends and data (de)serialization, pattern generation/checking, and thermal tuning (TT) circuits, and is 3D-integrated with the PIC via Cu-Cu hybrid bonding [6]. Rings are ordered largest-first for crosstalk reduction [7].

Each ring’s TT circuits (Fig. 1b) convert its drop port current to an 8b code (“read path”) that is passed to a digital locking algorithm, which sets the power delivered to a heater co-integrated with the ring (“write path”). The read path comprises a current mirror whose output is integrated across

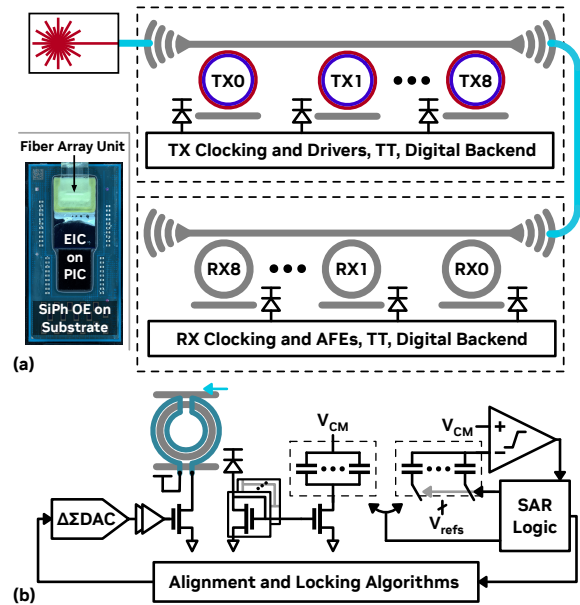


Fig. 1: (a) Test chip link; inset: packaged module. (b) TT diagram (V_{CM} , V_{refs} : ADC common-mode, reference voltages).

one of two ~ 0.5 pF capacitor banks; an 8b successive-approximation register (SAR) analog-to-digital converter (ADC) uses these capacitors in an alternating fashion to digitize the resulting voltage (one bank is used for integration while the other is used for the SAR process). The write path consists of a 13b delta-sigma ($\Delta\Sigma$) digital-to-analog converter (DAC) with pulse density-modulated output; the heater thermal pole averages the pulsed heater power into a temperature change that varies roughly linearly with the DAC code. Once the loop is locked, the algorithm continually adjusts the heater power to achieve either a target (TX) or maximum (RX) drop port current.

Static Ring Assignment

The link is first connected (laser to TX to RX) and powered up; each TT loop is unlocked while the read and write paths remain on, and all heaters are turned off. To begin, ring TX0 is swept from warm to cool and the drop current is recorded at each heater step (Fig. 2a). This sweep is used to identify lasers within range for TX0, which appear

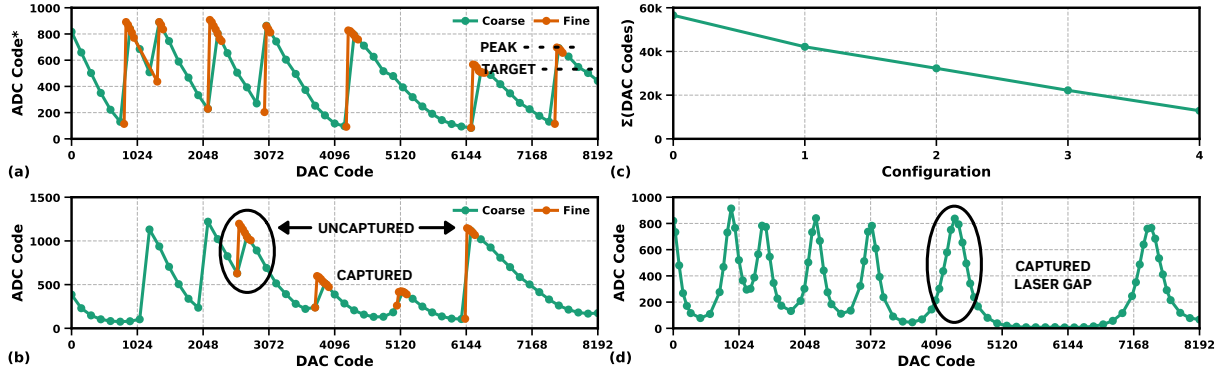


Fig. 2: (a) TX SRA, initial TX0 warm-to-cool heater sweep (coarse to find peak currents; then fine to find peak currents; all data are measured). In first configuration, redmost laser is chosen; target ADC code is set to 75% of peak current from laser. *8b ADC code is averaged to return an 11b code plotted here. (b) TX SRA, configuration 3, TX2 heater sweep; the ring is assigned to the redmost uncaptured laser (circled) blue of the bluemoost captured laser. (c) Sum of heater DAC codes vs valid configuration for TX SRA; configuration 4 is power-optimal (1k codes $\approx$ 3.3 mW). (d) RX SRA, configuration 1, RX2 heater sweep; the ring is assigned to the circled laser after confirming via upstream ring unlocking that the labeled gap corresponds to a captured laser.

as peaks; TX0 is then initially assigned to the redmost (highest heater power) laser within range. TX ring “assignment” means the ring is heated until its resonance lies red of the laser, returning it to the correct bistable state with respect to that laser; then, the ring’s TT loop is turned on with a target equal to $\sim 75\%$ of the peak current seen by the ring from that laser, which maximizes the optical modulation amplitude. As shown below, locking the ring to its laser ensures that no downstream ring locks to the same line; it also keeps the locked ring from being disturbed by heater crosstalk from downstream ring alignment.

Starting with TX1 and going down the bus, each ring is swept from warm to cool to identify lasers in range. If a current peak appears smaller in magnitude and/or less sharp than other peaks, it means the power at that line is smaller, which for uniform laser line power implies that an upstream ring is locked to that line, i.e., that laser is “captured” (Fig. 2b). Thus, peak magnitude and sharpness are used to classify each laser as captured or uncaptured. After a ring is swept, it may see up to two groups of consecutive captured lasers. The ring is assigned to the redmost uncaptured laser blue of the bluemoost captured laser (in the redder of the two captured groups, if the ring sees two groups). This ensures that the target ring is locked to the laser that is just blue of the line to which the preceding ring is locked – and thus enforces that the spectral order of the rings matches their spatial order, up to a rotation. If a ring sees only one group of captured lasers with no uncaptured laser blue of them, then it either locks to the redmost uncaptured laser (if TX0 is locked in the upper half of its range), or that configuration is deemed invalid (if otherwise). If a ring does not see captured lasers at all, it is assigned to the redmost laser within range.

Once all TX rings are assigned, their heater DAC codes are summed and stored. The process

is then restarted, but with TX0 assigned to the second-redmost laser in range. In this way, every possible configuration is tested and the one with minimum heater power is chosen (Fig. 2c). Depending on ring and laser locations and the time it takes to rule out invalid configurations, one may choose to have TX0 lock to lasers from blue to red instead for shorter total alignment time.

Once TX rings are assigned in the minimum heater power configuration, RX assignment commences. The major difference on the RX side is that for critically coupled RX rings properly locked to their assigned lasers, downstream rings do not see captured lasers at all – instead, they see no drop current at the corresponding heater power (Fig. 2d). When assigning downstream rings, searching for gaps between drop current peaks is used to screen for captured lasers, but this is not sufficient to rule out gaps due to nonuniform laser spacing or an unfilled ring free spectral range. To do so, when a candidate gap is identified, the immediately upstream ring is unlocked and heated, and then the target ring is re-swept over that gap; if a laser now appears in that gap, then the target ring is assigned to the redmost laser blue of the gap, and the upstream ring is re-locked to its laser. This method may also be used on the TX bus to make TX ring assignment more robust to laser line nonuniformity. RX ring “assignment” means the ring is heated to the chosen laser and the ring’s TT loop is turned on in current-maximization mode. Besides this method of identifying captured lasers, RX SRA proceeds in the same way as on the TX bus; once it completes, we proceed with measuring BER bathtub curves and then enable continuous link operation [4,5].

SRA as presented here is more robust to laser and ring variation than the method presented in [8], working even in the presence of variation-induced ring resonance swaps. It is similar to that in [9,10], but it is free of the requirement that all

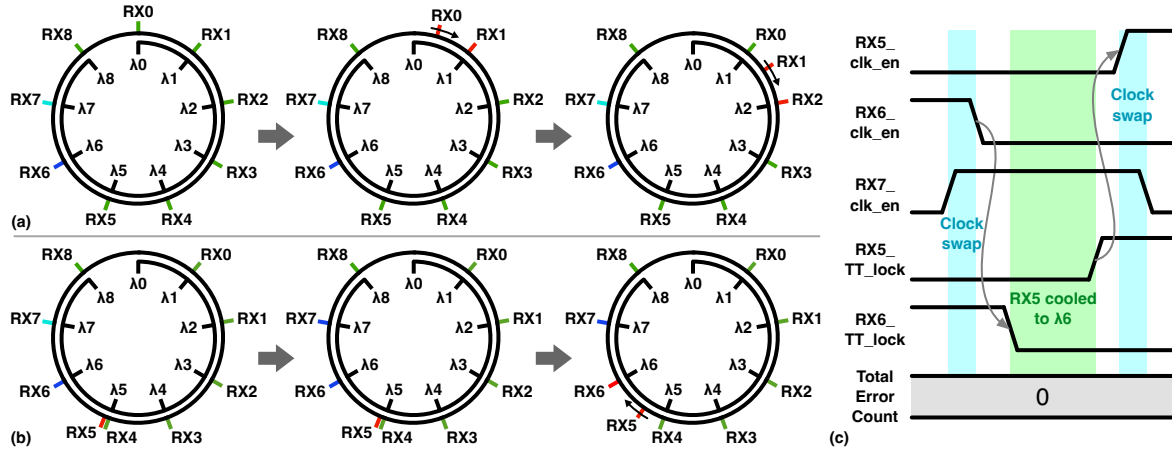


Fig. 3: In (a) and (b), inner circles depict lasers (clockwise ↔ bluer/cooler) and unbroken outer circles depict periodic RX ring resonances. Ring marker legend: green ↔ data, red ↔ TT unlocked, dark blue ↔ used clock, light blue ↔ spare clock. (a) Start of DRA. Left: Pre-DRA, the link starts with 7 data and 2 clocks. Center: RX1 TT frozen; RX0 cooled to λ_1 . Right: RX0 locked to λ_1 ; the process continues with RX1/RX2. (b) Rotation with clock swap. Left: RX4 has cooled and locked to λ_5 ; RX5 TT frozen. Center: RX7's spare clock is used in conjunction with RX6's clock to drive RX clock distribution. Right: RX6 TT frozen; RX5 cooled to λ_6 . (c) Link remains error-free during clock swaps (based on measured error count).

upstream rings be detuned from lasers, making it more robust to choices of channel spacing, ring Q, number of rings on the bus, thermal crosstalk, and especially self-heating, which “smears out” resonances to the point that rings may never be completely detuned from nearby lasers, regardless of the rings’ heater codes (Fig. 2a).

Dynamic Ring Assignment

During link operation, if chip temperature falls sufficiently, it may be possible to rotate the ring mapping to arrive at a lower-heater power configuration that still maps each ring to a laser in such a way that the link suffers only momentary partial downtime, even for a clock-forwarded system. We call this dynamic ring assignment (DRA) and demonstrate it with an RX blueward-by-1 rotation.

For this demo, we configure the 9 λ link as seven data and two clock lanes (one clock drives the RX clock distribution network; the other is a spare). In practice, before DRA, one data lane would be configured as a spare clock lane, and after DRA, the spare clock lane would be reconfigured as a data lane. This is necessary because the clock cannot be simply taken offline, even temporarily, like the other data lanes can. To begin, we unlock both RX0 and RX1 from their lasers and freeze their heater codes; then, we cool RX0 until it finds the first laser blue of its current laser and lock it to that newly found line (Fig. 3a). RX0 is thus locked to λ_1 , the line to which RX1 was previously locked. Due to chromatic dispersion and path length mismatch, the received data phase from the λ_1 channel relative to the clock may differ post-rotation. In this test chip, signal phase is adjusted at the TX, so we modify the phase of the TX lane locked to λ_1 based on post-SRA calibration to ensure these phases

align post-rotation. Then, we restart pattern checking at RX0, bringing that lane back online.

Next, we unlock RX2 and cool RX1 to lock to RX2’s laser. This process continues until all rings RX(n) are locked to lasers $\lambda(n+1)$, with RX8 locking to λ_0 . Because SRA matches spectral and spatial ordering (up to a rotation) and is the starting point for DRA, moving each ring to the next-blue line during DRA preserves this ordering. During DRA, at most two data lanes are down at once, in addition to one more used as a spare clock; this combined with total rotation time, t_{rot} , impacts how often DRA may be triggered in response to chip temperature changes. In this demo, t_{rot} is limited by scan chain latency, but in principle it is roughly the number of rings multiplied by the time it takes to move one ring by one channel spacing. With DRA implemented in hardware and for a rotation by one 200 GHz channel spacing ($\Delta T \sim 16K$), we estimate $t_{rot} < 1$ ms.

Special care must be taken when the ring being moved or the ring whose laser is being captured is receiving the clock driving the RX clock distribution network (Fig. 3b). In either case, before moving the ring, we swap the used clock with the spare in a make-before-break manner; the data channels remain error-free during this “clock swap,” demonstrating the success of the 16 GHz clock handoff (Fig. 3c). Our DRA approach may be extended to clock-embedded DWDM links by simply eliminating the clock swap steps.

Conclusion

We have demonstrated SRA and DRA in a 9 λ clock-forwarded DWDM link. The former provides *in-situ* initialization with spectral-to-spatial order matching at minimum heater power, while the latter maintains this matching while reducing heater power as chip temperature drops.

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